

# Dev Sanghvi

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## Education

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| <b>Rice University</b><br><i>Master of Computer Science (AI), GPA 3.69/4.0; Teaching Assistant, Spring 2026</i> | Aug 2025 - Aug 2026<br>Houston, TX      |
| <b>Adani University</b><br><i>B.Eng. in Computer Science and Engineering (AI-ML), CPI 9.34/10</i>               | Sep 2021 - Apr 2025<br>Ahmedabad, India |

## Experience

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| <b>Co-Founder</b><br><i>DHI Technologies, Inc.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Nov 2025 - Present<br>Houston, TX       |
| <ul style="list-style-type: none"><li>Built DHI's video analytics engine (Python, FastAPI): <b>31</b> configurable use cases, including license plate recognition, long-term re-identification, intrusion, loitering and tailgating, on <b>12</b> detector engines.</li><li>Designed the multi-camera inference pipeline on a shared TensorRT context and shared-memory frame buffer: six live cameras in <b>710 MB</b> of memory on an 8 GB edge GPU, down from about <b>3.4 GB</b>.</li><li>Built the multi-tenant cloud dashboard (React, TypeScript, Cloudflare Workers, D1) with live alerts and webhooks.</li><li>Lead DHI Labs, the company's 12-product applied research program, with one tested repo per product.</li></ul> |                                         |
| <b>Team Leader</b><br><i>Adiance Technologies Pvt. Ltd.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Jan 2025 - Jul 2025<br>Ahmedabad, India |
| <ul style="list-style-type: none"><li>Built real-time activity tracking for edge video analytics with transformer models (Florence-2, TimeSformer).</li><li>Wrote a C++ ONNX Runtime face verification module for on-device authentication and tuned its latency.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                         |
| <b>AI Solutions Developer</b><br><i>VMukti Solutions</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Jul 2021 - Jan 2025<br>Ahmedabad, India |
| <ul style="list-style-type: none"><li>Built anomaly detection and activity tracking analytics running across <b>10K+</b> IoT cameras on a cloud VMS.</li><li>Compressed models for edge devices: <b>60-80%</b> smaller and <b>2-5x</b> faster at inference with accuracy held.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                         |
| <b>Machine Learning Researcher</b><br><i>Indian Space Research Organisation (ISRO)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Jan 2025 - Apr 2025<br>Ahmedabad, India |
| <ul style="list-style-type: none"><li>Built an offline OCR + transformer email threat filter: <b>98%</b> accuracy, about <b>70%</b> fewer false positives than rules.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                         |

## DHI Labs Research

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| <b>Edge scene graphs:</b> explainable alerts from any detector in <b>2.4 ms</b> per frame; R@20 <b>0.500</b> vs <b>0.220</b> baseline |
| <b>Cross-camera identity:</b> follows people across cameras with <b>zero</b> false matches (WILDTRACK, 313 people)                    |
| <b>Open-vocabulary detection:</b> adds object classes on-device from a text prompt, with <b>no</b> loss on old ones                   |
| <b>Occlusion-aware counting:</b> counts people the detector misses, with calibrated ranges; <b>34%</b> lower error                    |
| <b>Predictive alerting:</b> warns of incidents seconds ahead and audits every forecast; <b>8%</b> false alarms over 210 scenarios     |
| <b>Fixed-camera 3D:</b> turns a fixed camera into a metric sensor by calibrating from passers-by; <b>16 cm</b> position error         |
| <b>Prompt2Model:</b> text prompt to a compressed edge model; blocks any model below its accuracy floor                                |

## Publications and Other Research

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|------------------------------------------------------------------------------------------------------------|-----------|
| <b>Vision Transformer for Accelerated Metasurface Absorber Design</b> , CASML 2024, IISc (published)       | 2024      |
| <b>DHI whitepapers:</b> privacy-native edge video analytics; hybrid edge-cloud VMS architecture            | Jul 2026  |
| <b>Research Intern, IIT Bombay:</b> IoMT intrusion detection, <b>98%</b> accuracy on 77 GB of traffic data | 2024      |
| <b>Rice:</b> collision-risk forecasting with spatiotemporal GNNs (led models); LLM faithfulness            | 2025 - 26 |

## Technical Skills

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| <b>Languages:</b> Python, C++, TypeScript, JavaScript, SQL, C, Go                                                |
| <b>ML and vision:</b> PyTorch, TensorRT, ONNX Runtime, OpenCV, Hugging Face, QAT/PTQ, GNNs, conformal prediction |
| <b>Edge and video:</b> NVIDIA Jetson, GStreamer, RTSP/ONVIF, MediaMTX, Docker, Linux                             |
| <b>Backend and web:</b> FastAPI, React, Cloudflare Workers and D1, PostgreSQL, SQLite, Ollama (local LLMs)       |

## Awards and Leadership

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| Rice Summer Venture Studio (Liu Idea Lab), \$15,000 award; AI Venture Velocity Challenge semifinalist (top 27) | 2026      |
| 1st Place, e-Yantra Robotics (IIT Bombay); AU-IQM Hackathon winner; Vice President, ASPDC                      | 2022 - 24 |